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An adaptive wavelet-Galerkin method for an elastic-plastic-damage constitutive model: 1D problem

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Abstract

We present a wavelet-Galerkin method for the analysis of elastic-plastic-damage systems. Ju's model (J.W. Ju, Int. J. Solids Struct. 25 (1989) 803) is explicitly considered and numerically tested but the method applies equally well to a broad range of damage-mechanics models. A three-step elastic-predictor, plastic-corrector, damage-corrector strategy is the core of the proposed approach. We use a displacement approach with wavelet interpolating functions instead of classical polynomials. This allows to clearly locate those sites in the system where damage is taking place, thanks to the high-pass filtering action operated by the wavelets. This opens the way to inexpensive self-adaptive strategies that are introduced and preliminary tested herein. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Damage and fracture mechanics; Plasticity; Multiple scaling

1. Introduction

In recent years, in particular in the past three decades, damage mechanics has emerged among more classical theories for the mechanical constitutive responses of engineering materials. The goal is the description of distributed material damage including material stiffness degradation and microcrack initiation and growth. Thermodynamics teach how to model damage by considering irreversible processes involving energy dissipation and stiffness variation due to physical changes in microstructure.

As pointed out in [18], there are at least three different levels of scale of 'damage' in material responses. The first is related to the atomic voids and crystal defects and requires non-continuum mechanics models at the atomic scale. The second is related to microcracks and microvoids and requires micromechanical damage models or phenomenological continuum damage models. The third involves macrocracks and requires fracture mechanics models to model the growth of discrete macrocracks [19].

Damage mechanics has been applied to model creep damage, fatigue damage and creep–fatigue interaction, elasticity coupled to damage, plastic behavior for ductile and brittle materials such as concrete and rock [27,28] and nonlinear elastoplasticity coupled with distributed interacting microcracks. Continuum damage mechanics is based on the thermodynamics of irreversible processes, the internal state variable theory and relevant physical issues such as homogenization, kinetic law of damage growth and plasticity-damage coupling mechanisms. A scalar damage variable is suitable for characterizing isotropic damage processes. However, a tensor-valued damage variable (fourth order) is necessary in order to account for

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anisotropic damage effects [17]. An excellent survey on strain localization (along with several original results) may be found in [26].

The enormous interest of the scientific community for localization phenomena, which is in minimal part outlined herein, has so far mainly been focused on thermodynamics based material models, see also [2,3], in an internal variable framework [27,28]. From a physical and numerical point of view, one of the key problems of localization is the loss of ellipticity (or hyperbolicity in the dynamic case) in the governing equations that occurs in correspondence to the onset of strain localization. This phenomenon is closely related to the inherent scale multiplicity of the problem. Regularized models have therefore been proposed that belong to the following categories:

- gradient-dependent softening plasticity, see, e.g., [11];
- cosserat-continuum models, see, e.g., [10];
- nonlocal models, see, e.g., [6].

All these remedies are in fact based on the introduction of a material length-scale that is roughly speaking related to the width of the strain localization band and is some order of magnitude smaller than the overall scale of the structure. From a computational viewpoint, the most successful approach available so far seems to be the adaptive finite-element scheme developed in [12,13,21]). The basic idea is to modify the original Zienkiewicz–Zhu strain-energy criterion [29] with new criteria, capable to handle the ellipticity loss. The indicator suggested by Loret is a combination of element velocity variation, equivalent plastic strain and elastoplastic to elastic acoustic tensor ratio.

As to analytical approaches, an appealing analytical theory is the one developed in the pioneering papers by Hughes and associates, see e.g., [14,16], that, respectively, present an abstract variational multiscale theory and a displacement-based multiple-scale formulation for composite damage 1D materials. Regularization is introduced via a dissipated energy criterion and nonlocal formulations. Theoretical results about the multiple-scale analytical approach are presented and numerically validated. Mesh invariant load-displacement curves, as opposed to mesh-dependent ones due to standard Galerkin FEM, are among the most important goals reached by means of the mutiple scale approach.

A mention of its own should be attributed to the family of mesh-free methods that have recently found application in the analysis of nonlinear systems with strain localization. The reproducing kernel method, see, e.g., [5] and references therein, and other similar methods described in [20] introduce an inherent length-scale into the formulation that allows the regularization of material instabilities in a more natural fashion than more classical nonlocal schemes such as the gradient approach. Numerical studies are produced to validate the theoretical framework that are concerned with one and two dimensional systems with plastic-damage constitutive law.

In view of the importance of scale multiplicity in localization problems, we present herein a numerical method for elastic-plastic-damage 1D systems that uses wavelets as interpolating functions, i.e., an inherently multi-scale basis. The main advantage of such an approach is the capability of wavelet coefficients to indicate regions where the displacement field lacks regularity, i.e., where refinement is needed. Refinement consists then in adding higher scale wavelets to enhance details over a rather coarse approximation. Interpolating-wavelet-based methods were previously developed and applied to the analysis of dynamic elastoplastic systems with hardening within collocation [25] and Galerkin frameworks [7]. Biorthogonal wavelets were used in [24], where an analysis of convergence with respect to the wavelet details level was performed.

The paper is divided into three parts. The first part presents the topics of wavelet analysis and introduces the concept of multiresolution. It is possible to prove that the size of wavelet coefficients gives information about the regularity of the analyzed function. The goals of such a multiscale method are adaptivity and capability of handling singularities. One of the main good features of wavelet basis consists in the characterization of the smoothness of a given function in terms of the coefficients of the transform. Sobolev and Besov norms of a function may be characterized solely in terms of its wavelet coefficients.

The second part reports the elastic-plastic-damage model, with the hypothesis on the material and some computational issues pertaining the numerical algorithm. The elastic-plastic constitutive law classically needs to be introduced in incremental form, since the stress does not only depend on the current strain (as it happens in the purely elastic case), but on the whole past strain history. A rate

independent elastic-plastic material with isotropic and ductile damage is considered. The constitutive equations are formulated in perfect analogy to plasticity models, with a damage threshold allowed to vary during the damage process. The concept of effective stress enables the plastic and damage internal mechanisms to be coupled with each other. As usual in computational inelasticity, the core of the numerical procedure is an incremental modified Newton–Raphson scheme that amounts to an elastic-predictor, plastic-corrector and damage-corrector strategy. The state of damage in the material is characterized by a criterion which states that damage is initiated whenever the energy release rate exceeds an initial energy threshold. The evolution equation of the scalar damage variable is expressed by a consistency parameter which defines damage loading-unloading conditions, according to the Kuhn-Tucker relations classically used for plasticity. Experimental validation of our damage model is presented by means of a pull-out test.

The third part of the paper presents the results of a few numerical investigations regarding the incremental analysis of a pulled bar whose resistance has been weakened in a few sections to trigger strain localization. At this stage, adaptive meshing is only simulated but strong evidence of the capability of our wavelet-based method to give rise to an self-adaptive strategy emerges from the numerical results. A forecoming paper will discuss the adaptive method in detail.

2. Multiresolution wavelet analysis

The motivations behind the use of wavelet bases for the numerical solution of partial differential equations may be best elucidated by means of the concept of multiresolution analysis. The idea is to interpolate an unknown (and irregular) field at a coarse level by means of so-called scaling functions. Any improvement to the initial approximation consists in adding 'details' (where needed) provided by new functions (wavelets) that are to be orthogonal to the scaling functions in order to provide as many new details as possible. It is then apparent that wavelet methods are well suited for multiscale problems such as the strain localization one or boundary layer problems occurring in several areas of mathematical physics. Things are made more precise by the concept of multiresolution that is introduced hereafter in $L^2(\boldsymbol{R})$ for historical reasons but is a well defined concept for nearly all functional spaces of interest over compact sets.

A multiresolution analysis of $L^2(\boldsymbol{R})$ is defined [8] as a sequence of closed subspaces V_j with the following properties:

- $V_j \subset V_{j+1}$,
- $f(x) \in V_j \Longleftrightarrow f(2x) \in V_{j+1}$,
- $f(x) \in V_0 \Longleftrightarrow f(x+1) \in V_0$,
- $\bigcup_j V_j$ is dense in $L^2(\boldsymbol{R})$; $\bigcap_j V_j = \{0\}$.

The existence of a scaling function $\Phi(x) \in V_0$ is postulated such that the set $\{\Phi(x-k)\}$ is a basis of V_0. Let then W_j denote a subspace complementing the subspace V_j in V_{j+1}, i.e., $V_{j+1} = V_j \oplus W_j$. Each element of V_{j+1} can be uniquely written as the sum of an element in V_j and an element in W_j, which contains the details needed to pass from an approximation at level j to an approximation at level $j+1$. A function $\Psi(x)$ is called a wavelet if the set of functions $\{\Psi(x-k)\}$ is a basis for W_0. For a given function $f \in L^2(\mathbf{R})$, one can compute the coefficients of the transform as

$$c_{j,k} = \int_R f(x) \cdot \Phi_{j,k}(x)\,\mathrm{d}x \quad \text{[scaling coefficients]}, \tag{1}$$

$$d_{j,k} = \int_R f(x) \cdot \Psi_{j,k}(x)\,\mathrm{d}x \quad \text{[wavelet coefficients]} \tag{2}$$

with the definitions $\Phi_{j,k}(x) = 2^{j/2} \cdot \Phi(2^j \cdot x - k)$ and $\Psi_{j,k}(x) = 2^{j/2} \cdot \Psi(2^j \cdot x - k)$. Therefore, the wavelet transform gives following decomposition of $f \in V_{j_0}$:

$$f = \sum_k c_{0,k} \cdot \Phi_{0,k} + \sum_k \sum_{0 \leqslant j \leqslant j_0} d_{j,k} \cdot \Psi_{j,k}. \tag{3}$$

2.1. Choosing scaling and wavelet functions: some guidelines

The scaling function $\Phi(x)$ and the associated mother wavelet $\Psi(x)$ are far from being unique and in fact several families of wavelets are available. For our scope, only compactly supported wavelets are of interest and, having in mind classical FEM approximations for the 1D Laplace equation, we have chosen $\Phi(x)$ to be the usual linear 'shape function' one would use in a finite-element framework, see Fig. 1. The problem of finding the associated mother wavelet has been solved in [15], where under the constraint of piecewise linearity, it is shown that the associated mother wavelet is the function $\Psi(x)$ still presented in Fig. 1. Furthermore, wavelet functions at the boundaries should be slightly modified in order to comply with Dirichlet or Neumann boundary conditions, see Fig. 2. In general, the choice of wavelet functions that best approximate the solution to a given problem is quite an hard task. Restricting our interest to mixed variational approaches, e.g., wavelet-Galerkin strategies, a key point is the possibility of constructing wavelet bases complying with the LBB condition [4] by means of the so called biorthogonal wavelets that were developed mainly by Dahmen and coworkers [8]. Applications of such bases to the analysis of dynamic elastoplastic systems may be found in [24]. From a computational viewpoint, we should observe that computing any entry of the stiffness matrix, i.e., computing integrals of (product of derivatives) of wavelets, is quite an hard task when biorthogonal wavelets are chosen. The referential paper [9] presents an algorithm suited for such computations that transforms the quadrature problem into an eigenvalue one. Such an approach was also used in [24] as implemented in the MultiScale Library (MSLib) made available by the Aachen school of Mathematics [1].

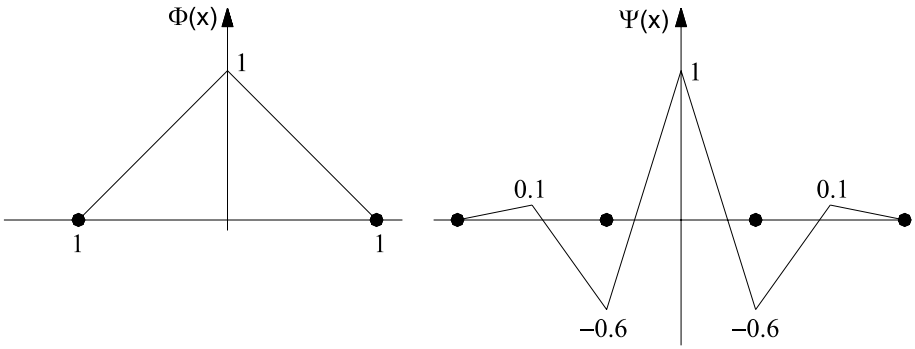

Fig. 1. Scaling and wavelet functions.

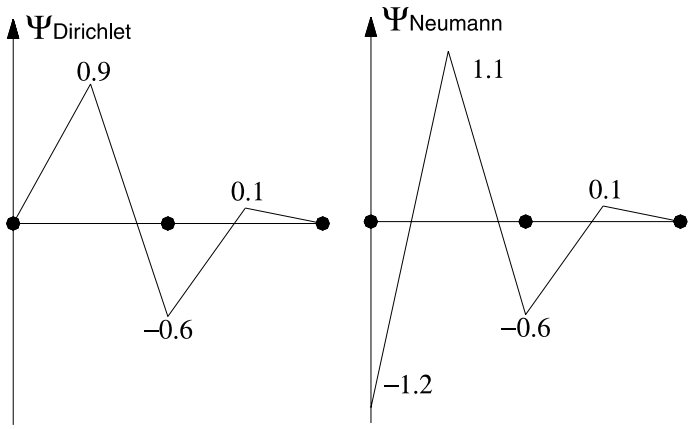

Fig. 2. Wavelet functions for Dirichlet and Neumann boundary conditions.

3. Material model: formulation and computational aspects

This section introduces the elasto-plasto-damage continuous model and its wavelet discretization. We will be using Ju's model [17] that is reported next.

3.1. The abstract problem

Let $u(x)$ be the displacement field, $\sigma(x)$ the stress field and $f(x)$ the axial load per unit length. Following [14], we give hereafter a rather formal variational formulation of the problem, without entering details about the functional spaces that give complete sense to the formulation itself. The domain Ω is an open connected subset of the real line whose boundary $\partial\Omega$ is the union of $\partial\Omega_u$, where displacements g may be imposed, and $\partial\Omega_t$ where tractions t may be applied. By posing $U = \{u \,|\, u = g \text{on } \partial\Omega_u\}$ and $S = \{\sigma \,|\, \sigma_n = t \text{ on } \partial\Omega_t\}$, the weak problem reads: find $u \in U$ and $\sigma \in S$ such that

$$\int_\Omega v_{,x}\,\sigma\,\mathrm{d}x - \int_\Omega vf\,\mathrm{d}x - \int_{\partial\Omega_t} vt\,\mathrm{d}S = 0 \quad \forall v \in V, \tag{4}$$

where $V = \{v \,|\, v = 0 \text{ on } \partial\Omega_u\}$ is the space of test functions and the unknown displacement field $u(x)$ is hidden into the stress field $\sigma(x)$. The following section is in fact devoted to elucidate the dependence of u on σ that is formally written as the history-dependent mapping $\sigma = N(\varepsilon, \alpha)$, where $\varepsilon = u_{,x}$ is the linearized strain measure and α is a vector of internal variables.

3.2. Formulation of the elasto-plastic-damage model

The total strain ε splits into the elastic ε^{e} and plastic ε^{p} parts. By obvious equilibrium considerations, the stress σ can be expressed by the elastic relationship

$$\sigma = E \cdot (\varepsilon - \varepsilon^{\mathrm{p}}), \tag{5}$$

where E is the Young modulus. Thanks to the flow rule, the rate of ε^{p} can be expressed for any admissible stress state σ as

$$\dot{\varepsilon}^{\mathrm{p}} = \gamma \cdot \mathrm{sign}(\sigma - q), \tag{6}$$

where $\gamma \geqslant 0$ is the amount of plastic strain rate and q is the additional internal variable of 'back stress', which defines the location of the center of the yield surface. The evolution of the back stress q is defined by the relation

$$\dot{q} = \gamma \cdot H \cdot \mathrm{sign}(\sigma - q), \tag{7}$$

where H is the so called isotropic hardening modulus. The strain-hardening plasticity leads to a yield criterion of the form

$$f(\sigma, q, \alpha) = |\sigma - q| - (\sigma_{\mathrm{y}} + K \cdot \alpha) \leqslant 0, \tag{8}$$

where $\sigma_{\mathrm{y}} > 0$ is the yield stress, K is the plastic modulus and α is the internal hardening variable, a non-negative function of the amount of plastic flow, whose evolutionary equation reads

$$\dot{\alpha} = \gamma. \tag{9}$$

The Kuhn–Tucker conditions express the physical requirement that the stress must be admissible and the plastic flow can take place only on the yield surface, i.e.,

$$\gamma \geqslant 0, \quad f(\sigma, q, \alpha) \leqslant 0, \quad \gamma \cdot f(\sigma, q, \alpha) = 0. \tag{10}$$

Finally, the consistency condition corresponds to the physical requirement that for $\dot{\varepsilon}^{\mathrm{p}}$ to be nonzero, the stress σ must persist on the yield surface, i.e.,

$$\gamma \cdot \dot{f}(\sigma, q, \alpha) = 0 \text{ if } f(\sigma, q, \alpha) = 0. \tag{11}$$

The model is now enriched through the introduction of a damage dissipation mechanism based on the concepts of damage mechanics that are briefly recalled next [19]. The term 'damage' relates to a class of micromechanical phenomena occurring during deformation of materials and producing degradation of the free energy. A damaged body is considered with a representative volume element at a point M oriented by a plane defined by its normal $\vec{n}$ and its abscissa x along the direction $\vec{n}$ (see Fig. 3). Let δS be the area of the intersection of the plane with the representative volume element and δS_{Dx} the effective area of the intersections of all microcracks or microcavities which lie in δS. The value of damage $D(M, \vec{n}, x)$ attached to the point M in the direction $\vec{n}$ at the abscissa x is written as

$$D(M, \vec{n}, x) = \frac{\delta S_{Dx}}{\delta S}. \tag{12}$$

In order to define a continuous variable over the representative volume element, one can consider the most damaged of all the planes varying with x, i.e.,

$$D(M, \vec{n}) = \max_x |D(M, \vec{n}, x)|, \tag{13}$$

so that the coordinate x disappears thus obtaining

$$D(M, \vec{n}) = \frac{\delta S_D}{\delta S}. \tag{14}$$

The value of the scalar variable D is bounded between 0 and 1, where $D = 0$ means undamaged material and $D = 1$ a fully broken one. For the one-dimensional case, damage can be defined as the effective surface density of microdefects, i.e.,

$$D = \frac{S_D}{S}. \tag{15}$$

The representative volume element of Fig. 4 is loaded by the force $\vec{F}$. The usual uniaxial stress is $\sigma = F/S$. It is convenient to introduce an 'effective stress' $\tilde{\sigma}$ related to the surface that effectively resists the load

$$\tilde{\sigma} = \frac{F}{(S - S_D)} = \frac{\sigma}{1 - D}. \tag{16}$$

An important issue in damage mechanics is the so called 'strain equivalence principle', which states that any constitutive equation for a damaged material may be derived in the same way as for a virgin material,

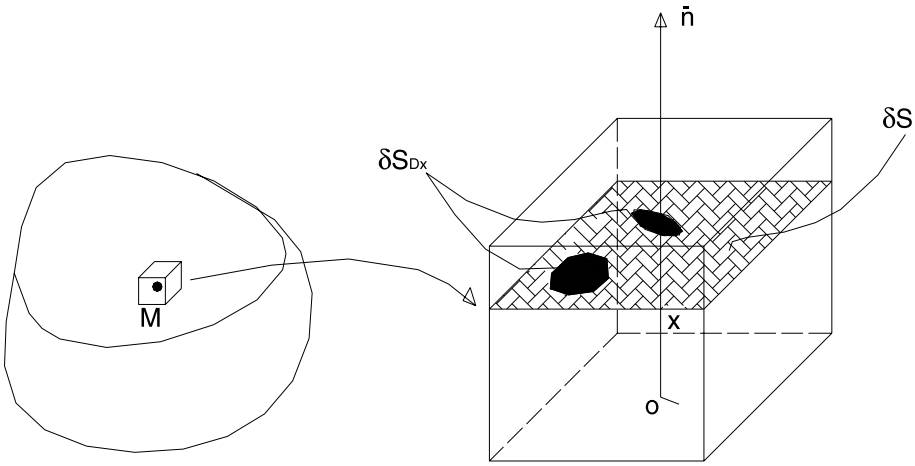

Fig. 3. On the definition of damage.

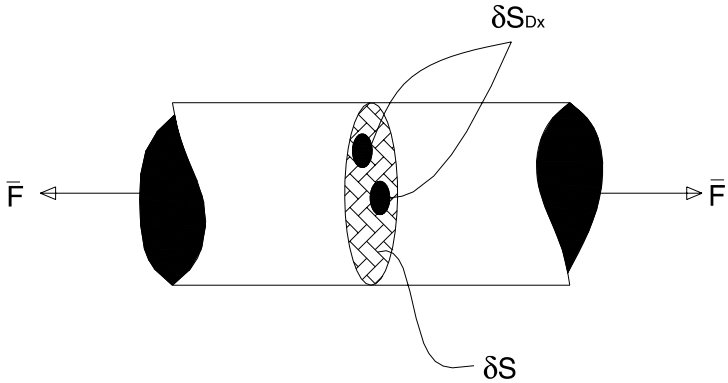

Fig. 4. One-dimensional damage.

except that the usual stress is replaced by the effective stress. To introduce both damage and plasticity mechanisms, the following free energy function is considered

$$\Psi(\varepsilon^{\mathrm{e}}, \alpha, \varepsilon^{\mathrm{p}}, D) = (1-D)\cdot\left(\tfrac{1}{2}\cdot E\cdot \varepsilon^{\mathrm{e}^2} + \tfrac{1}{2}\cdot K\cdot \alpha^2 + \tfrac{1}{2}\cdot H\cdot \varepsilon^{\mathrm{p}^2}\right). \tag{17}$$

The total potential energy function of the virgin material can be defined as:

$$\Psi^0(\varepsilon^{\mathrm{e}}, \alpha, \varepsilon^{\mathrm{p}}) = \tfrac{1}{2}\cdot E\cdot \varepsilon^{\mathrm{e}^2} + \tfrac{1}{2}\cdot K\cdot \alpha^2 + \tfrac{1}{2}\cdot H\cdot \varepsilon^{\mathrm{p}^2}. \tag{18}$$

Furthermore, the Clausius–Duhem inequality reads $\sigma\cdot\dot{\varepsilon} - \dot{\Psi} \geqslant 0$ whereas the stress and the effective stress may be evaluated as

$$\sigma = \frac{\partial\Psi}{\partial\varepsilon^{\mathrm{e}}} = (1-D)\cdot\frac{\partial\Psi^0}{\partial\varepsilon^{\mathrm{e}}}, \quad \tilde{\sigma} = \frac{\sigma}{1-D} = \frac{\partial\Psi^0}{\partial\varepsilon^{\mathrm{e}}}. \tag{19}$$

It is convenient to introduce the notation $\omega \equiv \Psi^0(\varepsilon^{\mathrm{e}}, \alpha, \varepsilon^{\mathrm{p}})$. The state of damage in the material at current time t is characterized by means of a damage criterion $g(\omega_t, r_t) \leqslant 0$, that simply reads

$$g(\omega_t, r_t) \equiv \omega_t - r_t \leqslant 0, \tag{20}$$

where r_t is the current damage threshold and r_0 denotes the initial damage threshold before any loading is applied, i.e., a property characteristic of the material. The equations of evolution for D and r_t may be written as

$$\dot{D}_t = \dot{\mu}, \quad \dot{r}_t = \dot{\mu}, \tag{21}$$

where $\dot{\mu} \geqslant 0$ is a damage consistency parameter which enters the damage Kuhn–Tucker conditions

$$\dot{\mu} \geqslant 0, \quad g(\omega_t, r_t) \leqslant 0, \quad \dot{\mu}\cdot g(\omega_t, r_t) = 0. \tag{22}$$

Eq. (22) implies that if $g(\omega_t, r_t) < 0$, the damage criterion is not satisfied and $\dot{\mu} = 0$, so no further damage is taking place. If conversely $\dot{\mu} > 0$ damage is accumulating and conditions (22) imply that $g(\omega_t, r_t) = 0$. The value of $\dot{\mu}$ is determined by the damage consistency condition that reads

$$g(\omega_t, r_t) = \dot{g}(\omega_t, r_t) = 0 \Rightarrow \dot{\mu} = \dot{\omega}_t, \tag{23}$$

with r_t given by the expression $r_t = \max\{r_0, \max_{s\in[0,t]}\omega_s\}$.

3.3. Computational issues

From a computational point of view, the introduction of damage in the classical rate independent plasticity Newton–Raphson scheme, needs the development of a 'damage corrector' after the usual 'elastic predictor' and 'plastic corrector'. The procedure allows to update the set of variables at the time $t = t_n$, i.e.,

$$\{\sigma, D, \varepsilon^{\mathrm{p}}, \alpha\}|_{t=t_n} = \left\{\sigma_n, D_n, \varepsilon^{\mathrm{p}}_n, \alpha_n\right\} \Rightarrow \{\sigma, D, \varepsilon^{\mathrm{p}}, \alpha\}|_{t=t_{n+1}} = \left\{\sigma_{n+1}, D_{n+1}, \varepsilon^{\mathrm{p}}_{n+1}, \alpha_{n+1}\right\}$$

as shown next.

3.3.1. Initialization

Having in mind the variational problem in Eq. (4), the method is started by solving an elastic trial problem whose unknowns are the coefficients of the wavelet decomposition of $u(x)$, see Eq. (3). The strain $\varepsilon(x)$ is then computed by differentiating $u(x)$ and the stress $\sigma(x)$ by the elastic rule $\sigma = E\varepsilon$. The plastic strain is initially set to zero, i.e., $\varepsilon^{\mathrm{p}} = 0$, and the internal variable vector α takes on values pertinent to the elastic domain.

3.3.2. Some remarks on the elastic trial problem

In the elastic range one obviously has

$$\sigma = E\varepsilon = Eu_{,x}$$

that, once substituted into the variational equation (4), gives rise to the well-known expression

$$\int_\Omega Eu_{,x}\, v_{,x}\, \mathrm{d}x - \int_\Omega vf\, \mathrm{d}x - \int_{\partial\Omega_t} vt\, \mathrm{d}S = 0 \quad \forall v \in V.$$

Recalling now that the displacement field $u(x)$ is expanded as a sum of scaling functions $\Phi(x)$ and wavelet functions $\Psi(x)$, that are also taken as test functions as in classical Galerkin frameworks, the stiffness matrix takes on the abstract form

$$K = EA \begin{bmatrix} \int_\Omega \Phi'_{j0,k}\Phi'_{j0,k} & \int_\Omega \Phi'_{j0,k}\Psi'_{j0,k} & \cdots & \int_\Omega \Phi'_{j0,k}\Psi'_{J,k} \\ \int_\Omega \Psi'_{j0,k}\Phi'_{j0,k} & \int_\Omega \Psi'_{j0,k}\Psi'_{j0,k} & \cdots & \int_\Omega \Psi'_{j0,k}\Psi'_{J,k} \\ \vdots & \vdots & \vdots & \vdots \\ \int_\Omega \Psi'_{J,k}\Phi'_{j0,k} & \int_\Omega \Psi'_{J,k}\Psi'_{j0,k} & \cdots & \int_\Omega \Psi'_{J,k}\Psi'_{J,k} \end{bmatrix}.$$

The presence of different wavelet levels at the same time makes the computations of the entries of the stiffness matrix quite an hard task. In fact, the elements of the stiffness matrix are of type

$$\int_\Omega c(x) D_1 g(2^n x - \alpha) D_2 h(2^m x - \beta)\, \mathrm{d}x, \tag{24}$$

where D_1 and D_2 are differential operators (first-order in our case), $c(x)$ is a possibly space-varying stiffness, n and m are scale parameters and α and β are shift parameters. It is apparent that ordinary quadrature formulae may fail when applied to integrals as above and ad hoc techniques should be devised on the purpose, see e.g., the one proposed in [9].

3.3.3. Elastic predictor

Given the displacement u_{n+1}, one can compute by differentiation the total deformation ε_{n+1}. The trial state is defined as follows:

$$\varepsilon^{p,\mathrm{trial}}_{n+1} = \varepsilon^{\mathrm{p}}_n, \quad \alpha^{\mathrm{trial}}_{n+1} = \alpha_n, \quad D^{\mathrm{trial}}_{n+1} = D_n,$$

$$\sigma^{\mathrm{trial}}_{n+1} = \left(1 - D^{\mathrm{trial}}_{n+1}\right) \cdot \frac{\partial \Psi^0\left(\varepsilon_{n+1} - \varepsilon^{p,\mathrm{trial}}_{n+1}, \alpha^{\mathrm{trial}}_{n+1}, \varepsilon^{p,\mathrm{trial}}_{n+1}\right)}{\partial \varepsilon^{\mathrm{e}}} = \left(1 - D^{\mathrm{trial}}_{n+1}\right) \cdot E \cdot \left(\varepsilon_{n+1} - \varepsilon^{p,\mathrm{trial}}_{n+1}\right),$$

$$\tilde{\sigma}_{n+1}^{\text{trial}} = \frac{\sigma_{n+1}^{\text{trial}}}{\left(1 - D_{n+1}^{\text{trial}}\right)} E \cdot \left(\varepsilon_{n+1} - \varepsilon_{n+1}^{p,\text{trial}}\right).$$

3.3.4. Plastic corrector

Check for yielding

$$f\left(\tilde{\sigma}_{n+1}^{\text{trial}}, \alpha_{n+1}^{\text{trial}}, \varepsilon_{n+1}^{p,\text{trial}}\right) \leqslant 0 \Rightarrow \text{elastic step} \Rightarrow \text{damage corrector},$$
$$f\left(\tilde{\sigma}_{n+1}^{\text{trial}}, \alpha_{n+1}^{\text{trial}}, \varepsilon_{n+1}^{p,\text{trial}}\right) > 0 \Rightarrow \text{plastic step} \Rightarrow \text{“return mapping”}.$$

Plastic ‘return mapping’ corrector

$$\boxed{f_{n+1}^{\text{trial}} \leqslant 0} \Rightarrow \begin{cases} \varepsilon_{n+1}^{\text{p}} = \varepsilon_n^{\text{p}}, \\ \alpha_{n+1} = \alpha_n, \\ q_{n+1} = q_n, , \\ \sigma_{n+1} = \sigma_{n+1}^{\text{trial}}, \end{cases} \qquad \boxed{f_{n+1}^{\text{trial}} > 0} \Rightarrow \begin{cases} \Delta\gamma = \frac{f_{n+1}^{\text{trial}}}{E+(K+H)} > 0, \\ \sigma_{n+1} = \sigma_{n+1}^{\text{trial}} - E \cdot \Delta\gamma \cdot \text{sign}\left(\xi_{n+1}^{\text{trial}}\right), \\ \varepsilon_{n+1}^{\text{p}} = \varepsilon_n^{\text{p}} + \Delta\gamma \cdot \text{sign}\left(\xi_{n+1}^{\text{trial}}\right), \\ q_{n+1} = q_n + \Delta\gamma \cdot H \cdot \text{sign}\left(\xi_{n+1}^{\text{trial}}\right), \\ \alpha_{n+1} = \alpha_n + \Delta\gamma, \end{cases}$$

where $\xi = \sigma - q$.

The state variables at the end of plastic corrector phase become $\{\tilde{\sigma}_{n+1}, D_n, \varepsilon_{n+1}^{\text{p}}, \alpha_{n+1}\}$.

3.3.5. Damage corrector

$$\omega_{n+1} \equiv \Psi^0\left(\varepsilon_{n+1} - \varepsilon_{n+1}^{\text{p}}, \alpha_{n+1}, \varepsilon_{n+1}^{\text{p}}\right), \quad D_{n+1} = \begin{cases} D_n & \text{if } \omega_{n+1} - r_n \leqslant 0, \\ D_n + (\omega_{n+1} - \omega_n) & \text{otherwise}, \end{cases}$$

$$r_{n+1} = \max\{r_n; \omega_{n+1}\}; \sigma_{n+1} = (1 - D_{n+1}) \cdot \tilde{\sigma}_{n+1}.$$

The final set of variables is $\{\sigma_{n+1}, D_{n+1}, \varepsilon_{n+1}^{\text{p}}, \alpha_{n+1}\}$.

As already pointed out in [17], plastic and damage correctors are (formally) uncoupled from an algorithmic point of view even though they are not so if one considers the model equations (17)–(23).

4. Experimental and numerical applications

4.1. Experimental validation

A comparison between a laboratory pull test and numerical results from the proposed wavelet-Galerkin model is presented first. The goal is to test the model with an experimental σ–ε curve obtained from a displacement control test. The data adopted for the material model are given in Table 1 and correspond to ordinary steel. For sake of completeness, we should mention that rigorous applications of system identification techniques have been proposed in the recent literature to extract parameter values form experimental data, see, e.g., [22,23] that were not however used herein.

Table 1
Numerical parameters for experimental/numerical validation

Cross-section	$A = 4.52$ cm^2
Young modulus	$E = 21000$ kN/cm^2
Plastic modulus	$K = 2000$ kN/cm^2
Isotropic hardening modulus	$H = 1500$ kN/cm^2
Yield stress	$\sigma_y = 70$ kN/cm^2
Initial damage threshold	$r_0 = 0.117$ kN cm

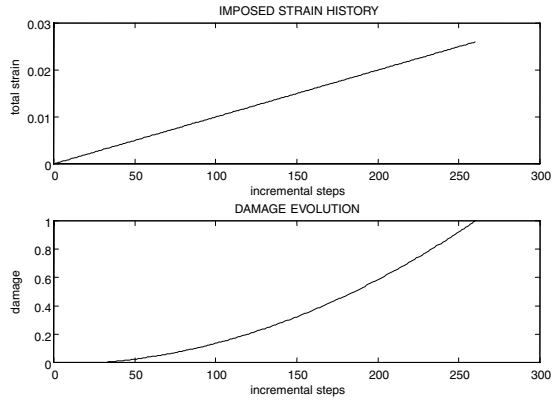

Fig. 5. Imposed strain history and damage evolution.

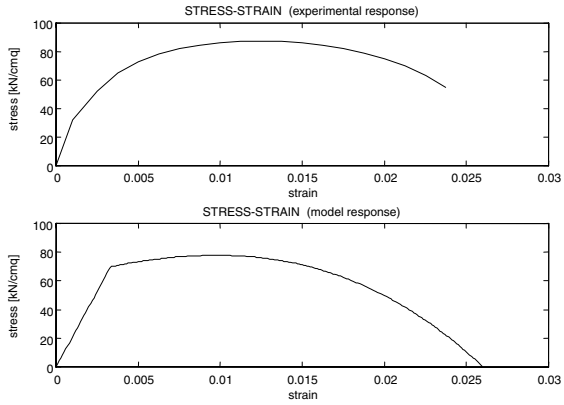

Fig. 6. Experimental and model σ–ε curves.

The imposed strain history and the damage evolution are shown in Fig. 5, while the experimental and model σ–ε curves are depicted in Fig. 6. It is important to remark that the falling branches in the σ–ε curve are due to the effect of the damage corrector phase and not to a plastic softening behaviour, so no arc-length procedure is employed in the return mapping algorithm. A good agreement is found except for values of D close to unity for which, as maybe expected, the numerical model becomes unnecessarily weak. The peak stress and the overall softening behavior are however well captured by the numerical model.

4.2. A numerical test

Another test for the algorithm is proposed next. A fully restrained bar subject to uniform load is considered as shown in Fig. 7. The resistance of the bar is uniform but for two sections that are weakened in order to trigger strain localization. In more detail, the parameters that fully define the numerical test are given in Table 2.

Figs. 8 and 9 show the space discretization of the bar enhancing the centers of the supports of scaling and wavelet functions at different levels. Damage onset is triggered in two sections, say A–A (leftmost section)

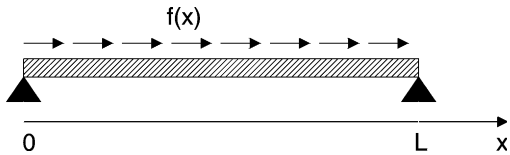

Fig. 7. Model problem.

Table 2
Structural parameters for the numerical test

Rod length	$L = 10$ cm
Cross-section	$A = 3.14$ cm^2
Young modulus	$E = 21000$ kN/cm^2
Plastic modulus	$K = 2000$ kN/cm^2
Isotropic hardening modulus	$H = 1000$ kN/cm^2
Yield stress	$\sigma_y = 20$ kN/cm^2
Initial damage threshold	$r_0 = 0.0482$ kN cm
Applied load	$f = 35$ kN/cm

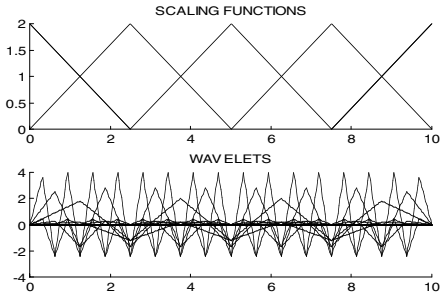

Fig. 8. Domain discretization.

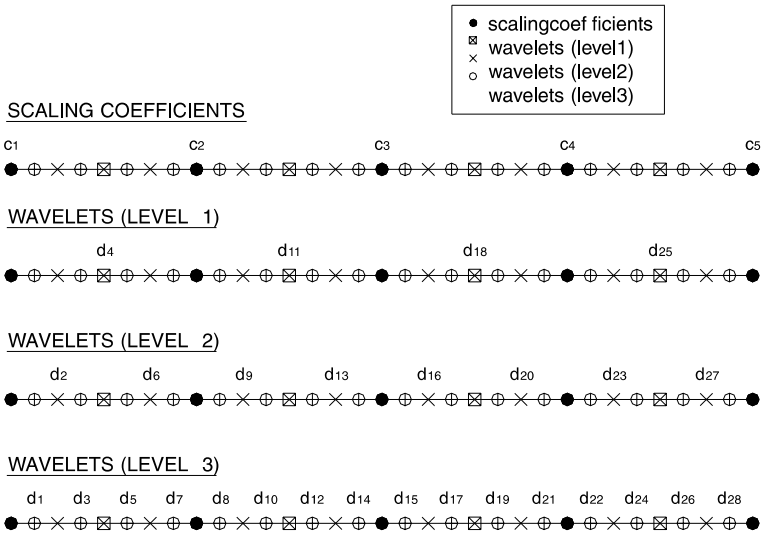

Fig. 9. Scaling and wavelet function support centre.

and B–B (one quarter of the total rod length) as shown in Fig. 10, with drops of material parameters as shown in Table 3.

Twenty steps were required to perform the overall analysis the details of which are discussed next. The displacement 3D plot (Fig. 11) shows how the first 10 incremental steps involve a perfectly elastic response (parabolic), while, in the second 10, the displacement function becomes flatter in the middle of the domain, because of plasticity and damage effects. The larger displacement increments occur in the last three loading steps, which present nonzero values of the damage variable. The final displacement in the middle of the rod has an increment of 44% as compared to the same problem with an elastic-plastic constitutive law without damage (0.036 cm instead of 0.025 cm). The lack of symmetry due to damage localization may be appreciated by looking at the numerical values of the displacement field. At the

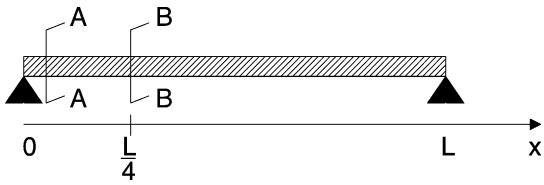

Fig. 10. Weakened sections.

Table 3
Weakened sections properties

A–A	$E = 18{,}900$ kN/cm^2	$K = 1800$ kN/cm^2	$H = 900$ kN/cm^2	$r_0 = 0.00482$ kN cm
B–B	$E = 14{,}000$ kN/cm^2	$K = 1333$ kN/cm^2	$H = 667$ kN/cm^2	$r_0 = 0.00482$ kN cm

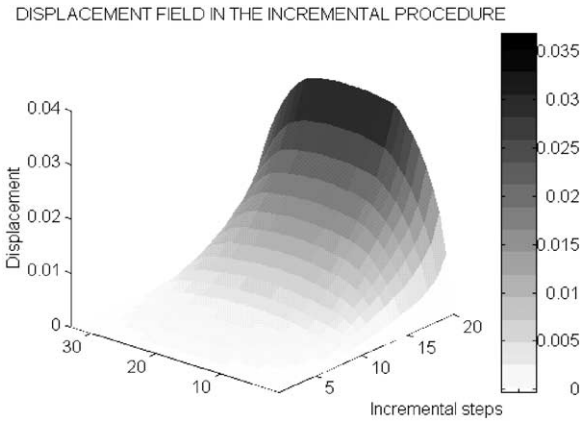

Fig. 11. Displacement along the rod vs iterations.

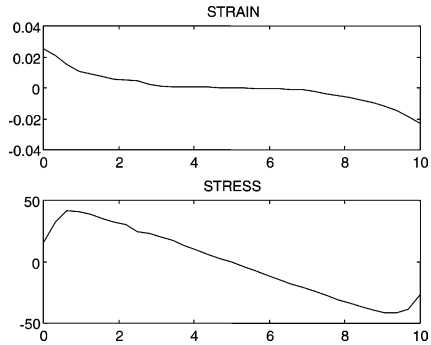

Fig. 12. Strain and stress along the rod at convergence.

leftmost section, the value of the total displacement is 0.0109 cm that is 18% larger then the rightmost value of 0.0092 cm.

Strain and stress curves all over the rod at the end of the incremental procedure are depicted in Fig. 12. The strain function is apparently consistent with the displacement function. Damage produces some discontinuities in the stress at the boundaries. This is consistent with the damage curve depicted in Fig. 15.

Fig. 13 shows the stress-strain curves for the scaling coefficient $c1$ and wavelet coefficients $d4$ (level 1), $d2$ (level 2) and $d1$ (level 3), all of which are positioned in the vicinity of the left boundary (see Fig. 9). Strain localization is made apparent by the fact that only the $c1$ and $d1$ curves have experienced the post-peak softening regime whereas the behavior of $d2$ and $d4$ is still an hardening one.

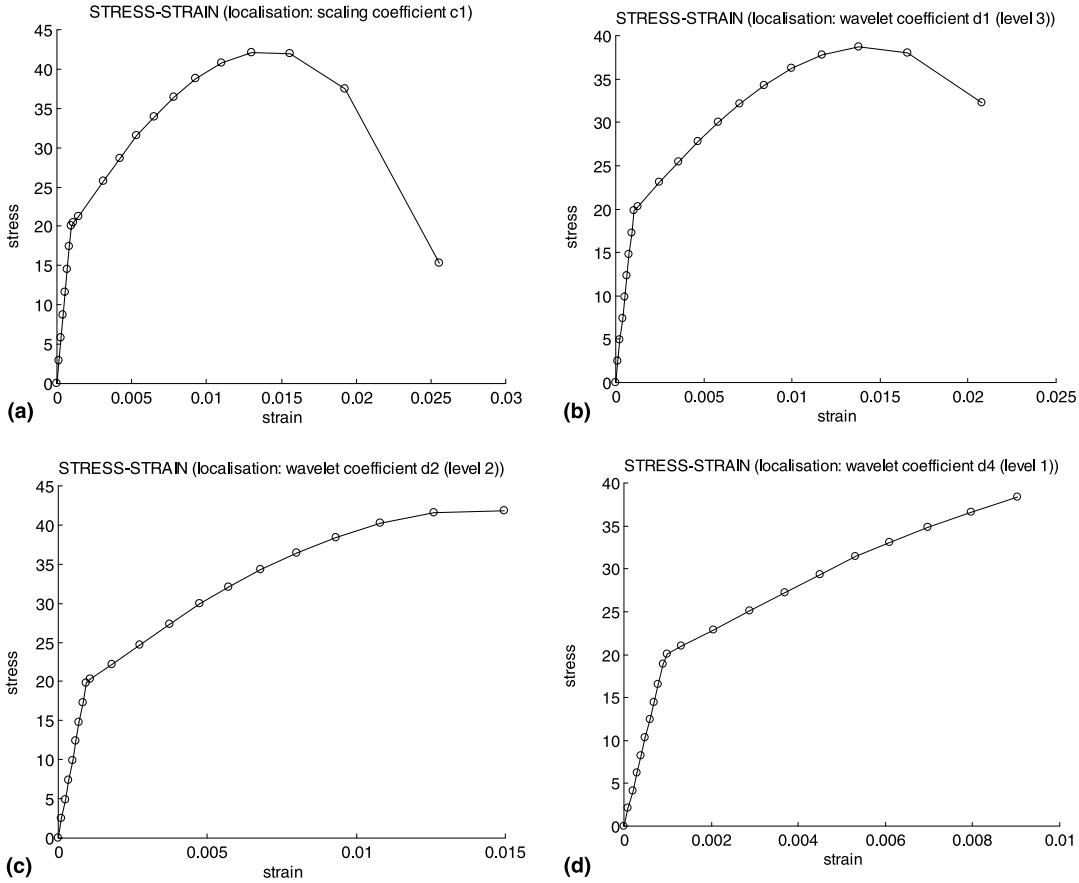

Fig. 13. Stress–strain curves for a few scaling and wavelet points close to the left boundary: (a) coefficient $c1$; (b) coefficient $d1$; (c) coefficient $d2$; (d) coefficient $d4$.

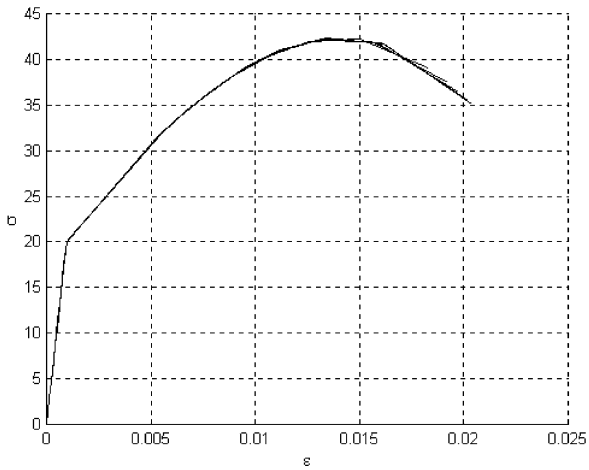

Fig. 14. Left boundary stress–strain for various wavelet refinements.

With the objective of investigating the objectivity of the results with respect to the spatial discretization, several analyses have been performed using 4 and 8 scaling functions and 4, 8, 16 and 32 wavelet functions. Fig. 14 shows the stress-strain curve of the leftmost section of the rod for all the cases. The solution is

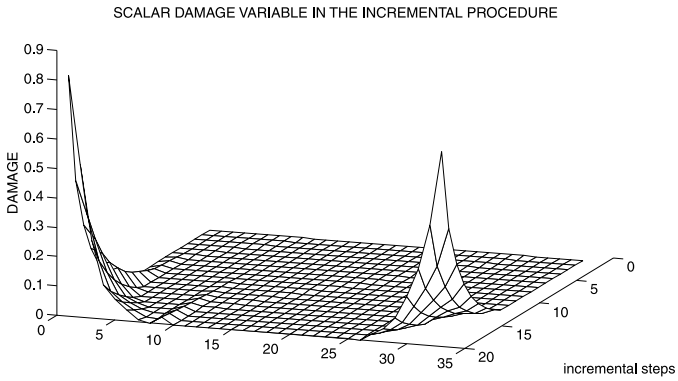

Fig. 15. Damage variable vs space and iterations.

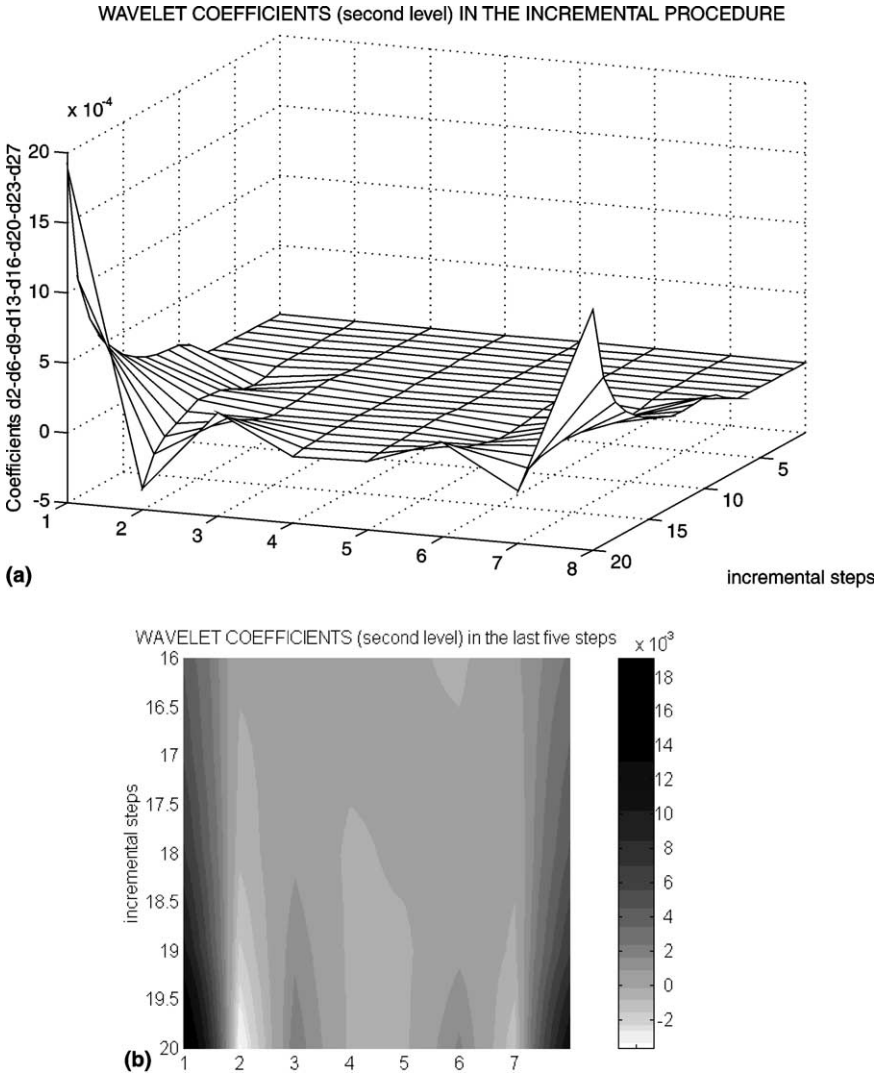

Fig. 16. Second level wavelet coefficients: (a) all iterations; (b) last five steps.

basically unique in the pre-peak region whereas a slight dispersion appears in the post-peak regime that seems however comparable with usual errors affecting all computational procedures.

The 3D plot of damage evolution with the incremental analysis is depicted in Fig. 15. Damage basically occurs at the boundaries where plastic flow is also localized. In fact damage is essentially related to the increment in the free energy of the system, which is a quadratic function of strain. In section B–B ($x = 1/4L$) one may see a discontinuity due to the reduced value of the initial damage threshold.

Figs. 16 and 17 show the evolution of the wavelet coefficients for the second and the third level. Three-dimensional plots as well as color maps (gray scale) of the same wavelet coefficients in the last five steps are shown. The capability of wavelet coefficients to locate the areas of damage localization is therein apparent. This is the main goal of the discretization method adopted.

4.3. Adaptivity

A further application is presented to prove the capability of the wavelet scheme to locate solution irregularities along the domain and automatically adapt the space discretization consequently. We consider a bar restrained on the left side and pulled by a force load on the right (Fig. 18). Damage is localized at the

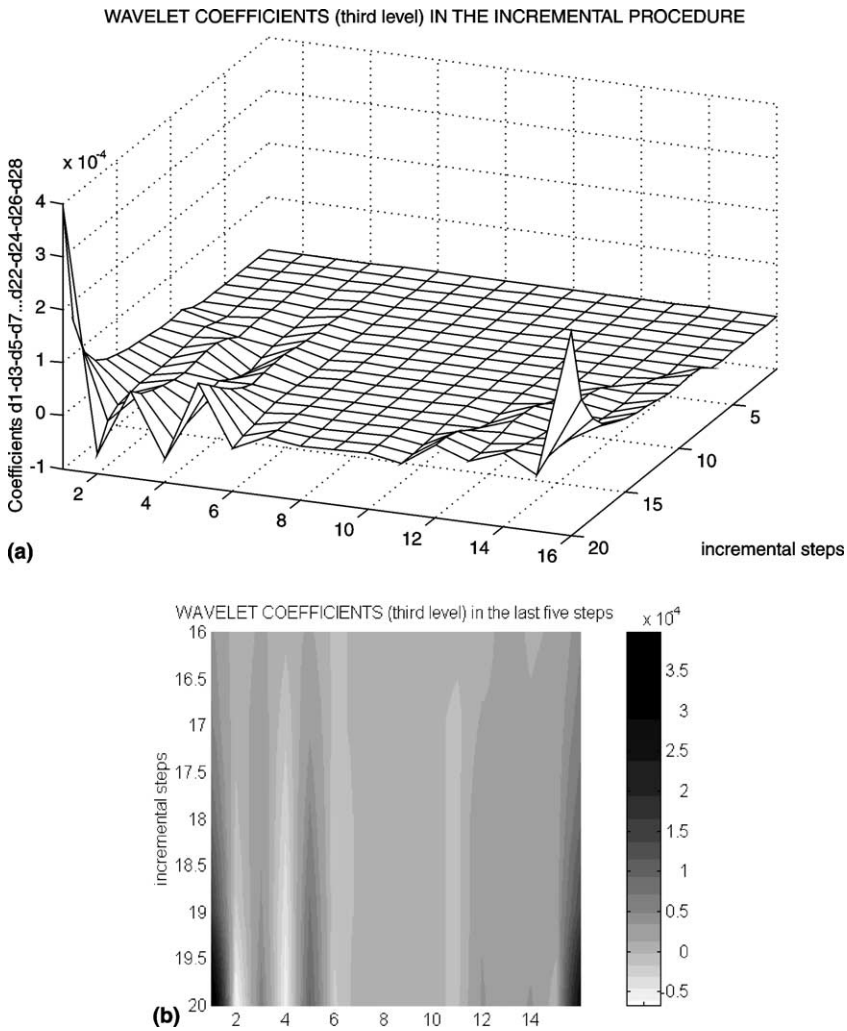

Fig. 17. Third level wavelet coefficients: (a) all iterations; (b) last five steps.

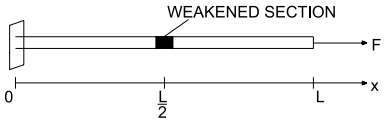

Fig. 18. Model problem.

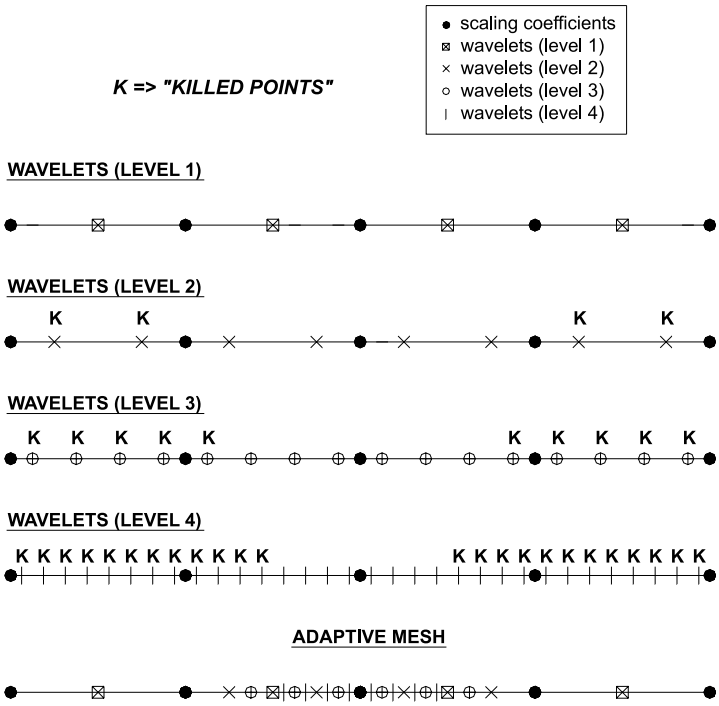

Fig. 19. Wavelet functions support centre – adaptive discretization.

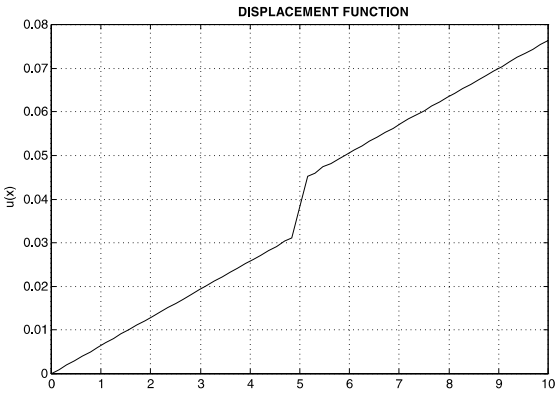

Fig. 20. Displacement field along the bar at convergence.

mid point of the bar by means of a 10% reduction of both the yield stress and the initial damage threshold. The incremental procedure is stopped as soon as the damage variable reaches one.

The idea is to start by defining a regular grid based on the regularity of the data, i.e., the load and the elastic stiffness of the rod, and solve the problem. The numerical solution is then analyzed looking at the

amplitude of the coefficients of the interpolating wavelets. Eventually, a new grid is defined by killing the points corresponding to small wavelet coefficients and adding new points in the vicinity of wavelets presenting large coefficients. Fig. 19 shows the evolution of the "mesh" along with the iterations, with four levels of wavelet details. In Figs. 20 and 21 the displacement field and the total strain at convergence are shown, while Fig. 22 shows the stress–strain curve at the middle point of the rod (Table 4).

The scalar damage variable is well localized in space in the central section (see Fig. 23). This is the proof which demonstrates that strain localization was successfully sparked off. Figs. 24 and 25 show the wavelet coefficients of the third and fourth level. The adaptive solution is the same as for the regular discretization, but the stiffness matrix size reduces from 64×64 to 28×28.

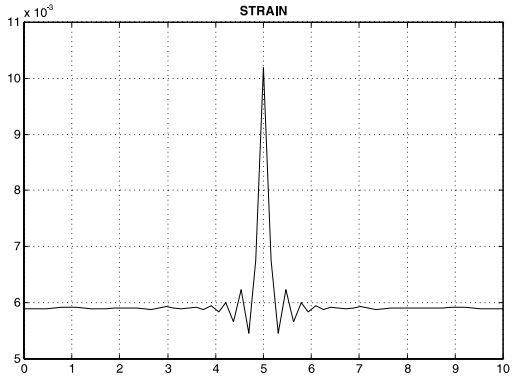

Fig. 21. Strain along the bar at convergence.

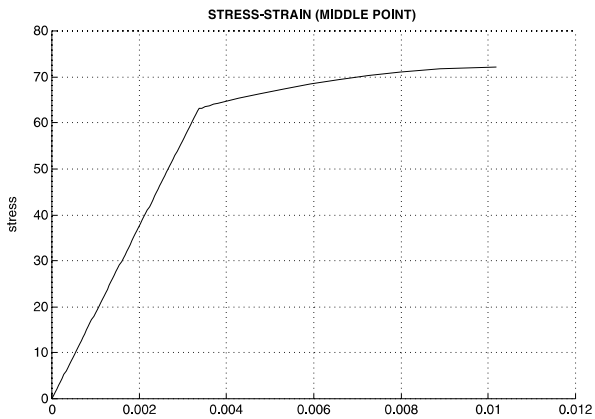

Fig. 22. Stress–strain curve at the middle point of the rod.

Table 4
Structural parameters for the numerical test

Rod length	$L = 10$ cm
Cross-section	$A = 2.01$ cm^2
Young modulus	$E = 18750$ kN/cm^2
Plastic modulus	$K = 2000$ kN/cm^2
Isotropic hardening modulus	$H = 1500$ kN/cm^2
Yield stress	$\sigma_y = 70$ kN/cm^2
Initial damage threshold	$r_0 = 0.13$ kN cm

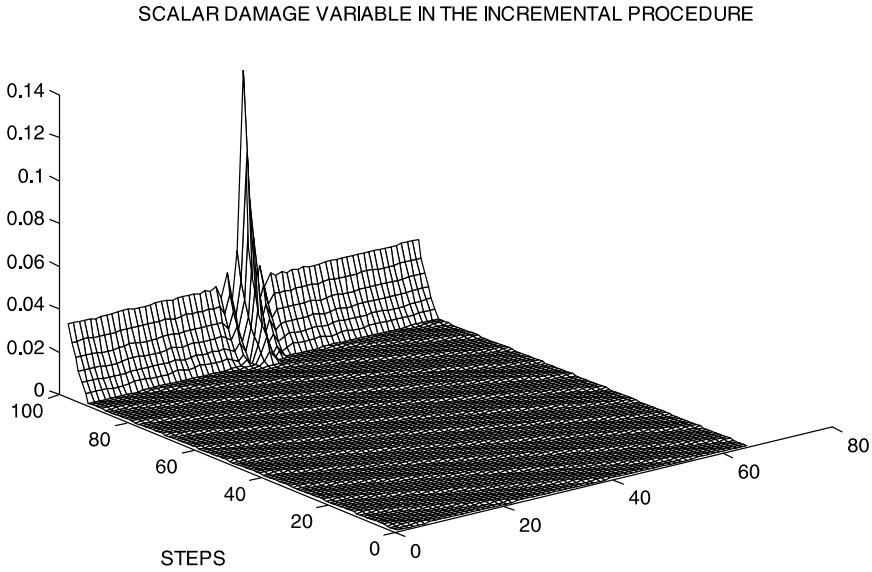

Fig. 23. Damage variable vs space and iteration.

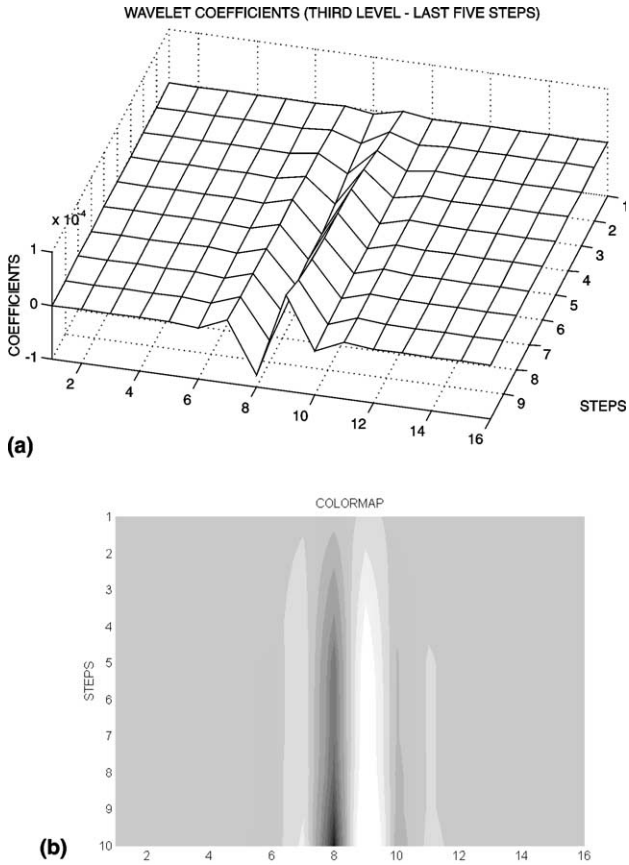

Fig. 24. Third level wavelet coefficients with gray scale map (last 10 steps).

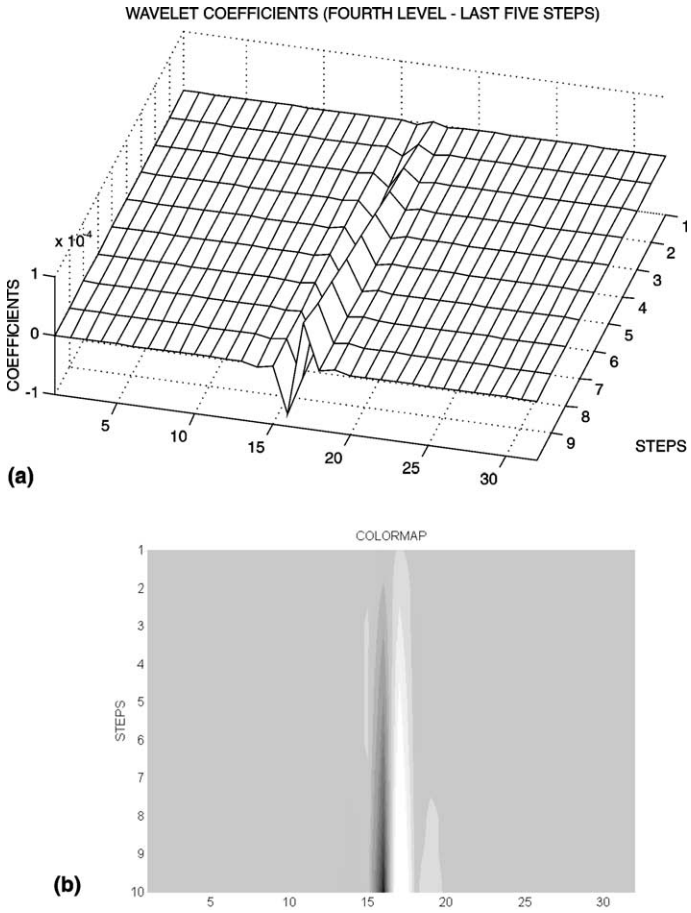

Fig. 25. Fourth level wavelet coefficients with gray scale map (last 10 steps).

5. Conclusions

We have introduced a wavelet-Galerkin method for the analysis of one-dimensional structural problems in the presence of an elastic-plastic-damage behaviour. The physical problem is a multiscale one as well as the wavelet basis introduced for its numerical solution. The inherent scale multiplicity of the adopted basis allows to clearly capture strain localization phenomena and opens the way to adaptive space discretization methods that do not require expensive computations of standard FEM error indicators. The proposed method was firs tuned and tested on the basis of experimental data regarding an ordinary steel bar whose behavior was satisfactorily replicated by the numerical model. Pathological numerical studies were then performed whereby the adaptivity of the method was outlined and preliminary investigated.

The use of other wavelet bases and the extension to multidimensional problems are the topics currently under investigation that seem to deserve further developments.

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